

Upload CSV

Choose a CSV File

std.csv

380.0B

Rule Engine Specification

The agent dynamically scans any dataset:

1. Universal Checks:
- o Duplicate rows.

o Null or missing values in any column.

o Mixed data types.
2. Text Formatting:
- o Outer whitespace warnings.
3. Numeric Outliers:
- o Mathematical outlier detection using the Interquartile Range (IQR).

o Negative checks on positive count fields.
4. Standard Constraints:
- o If `Age` is found: bounds checked [0, 100].

o If `email` is found: regex format validated.

o Missing checks in `name` and `roll number`.

API Secrets Configuration

Configure your OpenRouter API Key by creating:

- A `.streamlit/secrets.toml` file with `OPENROUTER_API_KEY="..."`

OR

A `.env` file in the root folder with `OPENROUTER_API_KEY="..."`

AI Data Quality Agent

Upload any CSV dataset to scan for data quality issues and anomalies. Our automated rule engine dynamically checks for missing values, duplicates, mixed types, email formats, and outliers, while **OpenRouter (Nex-AGI)** diagnoses patterns and writes custom Python cleanup scripts!

Dataset Preview

	student_id	name	class	math_score	attendance
0	101	Rahul Sharma	X-A	85.5	92%
1	102	Priya Singh	X-B	90	95%
2	103	Rohan Verma	X-C	105	102%
3	104	Sneha Gupta	10A	None	88%
4	105	Amit Yadav	X-B	-5	70%

Detected Column Attributes

Column Name	Inferred Type	Completeness (%)	Unique Values	Quality
student_id	Numeric (Integer)	100.0%	8	🟢
name	Categorical	100.0%	8	🟡
class	Categorical	100.0%	6	🟡
math_score	Numeric (Float)	75.0%	6	🟡
attendance	Categorical	100.0%	8	🟡
join_date	Categorical	100.0%	8	🟡

Detected Errors



AI Suggestions



⚠️ Detected 7 data quality warning(s)/anomaly(ies) in dataset:

- Column 'name', Row 2: Value contains leading/trailing whitespaces (' Priya Singh')
- Column 'class', Row 2: Value contains leading/trailing whitespaces (' X-B')
- Column 'math_score': Missing values at Row(s) 4, 8
- Column 'math_score': Negative value(s) found at Row(s) 5 (expected non-negative)
- Column 'math_score': Statistical outlier(s) detected: Row 5 (value: ...)

🔍 Diagnosis Summary

The anomalies are mainly caused by inconsistent data entry and weak validation at ingestion. Row 2 has leading/trailing spaces across multiple fields, which suggests copy/paste, manual entry, or export formatting issues; this can break matching, grouping, date parsing, and display consistency. The missing math_score values in rows 4 and 8 indicate incomplete score entry or extraction gaps. The negative math_score in row 5 violates the expected non-negative score rule and is also flagged as a statistical